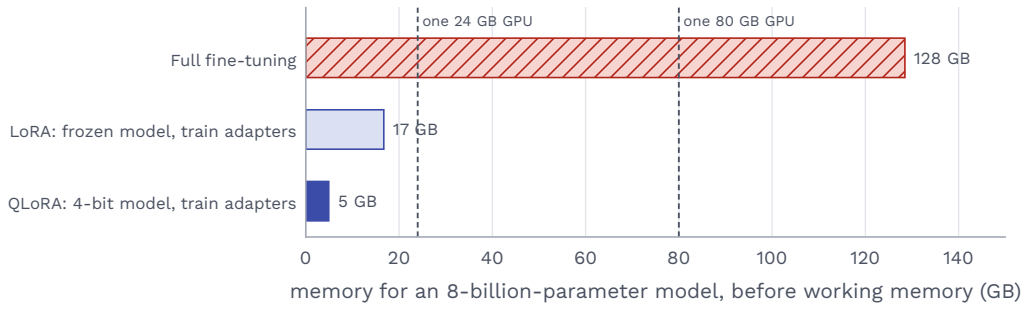
one 24 GB GPU
one 80 GB GPU
Full fine-tuning
128 GB
LoRA: frozen model, train adapters
17 GB
QLoRA: 4-bit model, train adapters
5 GB
0
20
40
60
80
100
120
140
memory for an 8-billion-parameter model, before working memory (GB)